

Emotions: shaping the future of everyday technology

A Hindu bride, beautifully adorned before her wedding, recently did something no other bride has ever done in India: Just before the ceremony, she chose to slip a small “Q sensor” on her leg, just above the ankle, hidden under her dress. The sensor recorded a signal known as skin conductance, an electrical measure on the surface of the skin that changes when innervations are received under the skin from the sympathetic nervous system. This system activates with emotional events, and such events were about to happen. The bride showed us her Q data leading up to the wedding: peaks when she was stressed (who wouldn’t be?) followed by a valley when her friends were chilling with her. The four biggest peaks came during the biggest emotional moments of the ceremony: First, during the solemn vows; second, when she moved to the right of the groom; third, when the groom tied the mangalsutra; and fourth, when they were declared husband and wife.

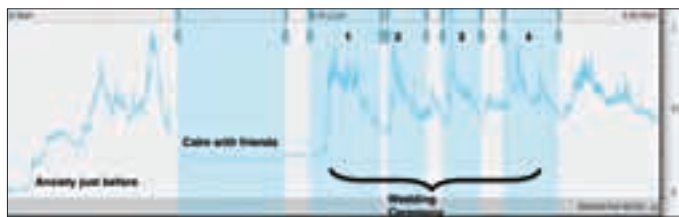
Creating technology to collect spontaneous emotional data, and to respond with sensitivity and respect to emotional events in the real world is a focus of my team’s research at MIT, and also now of a spinout company, Affectiva. The sensor worn by the bride is able to be used by scientists, doctors, and teachers for understanding and communicating stress – for example, in a boy who is autistic and non-speaking, who might experience hyper-sensitivity to sounds, lights or smells and be unable to describe this in words. When the boy wears the sensor and shows you his data, and you see it escalate with a particular sensory input, then you can debug what might be bothering him. Emotions are not always communicated accurately through face or voice. New emotion technology will enable better understanding. This same sensor, worn on the wrist, has sensitive

responses to seizures, and to many medications, and in some cases it might have a characteristic response to a drug craving or to a panic attack. In the

near future your cell phone will analyze patterns in signals from your body that you want to detect, and respond to those with customized support. It might bring you encouraging messages if you are struggling with stress-related overeating and if your sensor signals heightened stress. It could alert nearby people if you are having a seizure. It might crowd-source you an empathetic message when you’re feeling

down and need to feel hope restored.

When we set out to measure emotion, with engineering precision, we discovered that we were measuring physiological systems where changes in functioning almost always signaled something important. In the future, small sensors like the bride



chose to wear will be part of everyday life for people who wish to better understand and manage stress and medical conditions that are exacerbated by stress. The sensors will get smaller and cheaper and more fashionable with design. But the future of emotion technology is even greater.

How often have you been frustrated using technology, and when the next version comes out you learn they added a lot of features but did not fix the most frustrating parts of the experience? Did you make a facial expression of annoyance at that pop-up ad? But it was ignored. What if you actually liked an ad and smiled with delight after clicking on it, instead of looking bored? What if technology could see when it pleased or displeased users, transmitting only the anonymous expression if privacy is a concern, and let its creators know which parts were most frustrating? This “real world” feedback is starting to be part of our future: new software is enabling facial expressions to be communicated online, empowering users to send non-verbal messages to big companies, showing them exactly when customers are confused, upset, displeased, angry, annoyed, or interested and delighted. Affectiva online software by Affectiva is the first new step in this direction. Power to the people is coming in the online world – and companies are going to need to pay attention to this, just like they pay attention to you and your expressed emotions if you communicate face-to-face in their store. In the future, people will expect smart phones, tablets, online services, kiosks, computers, and robots, all to show respect for human feelings. 



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Learn more about the Sensor/Facial Emotion Recognition technology: www.affectiva.com